

ISE Industrial Computing – Homework Part 1

Custom MCP Math & Database Tool Server

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Public GitHub repository (clickable):

<https://github.com/markusrussold/ise-homework-part-1>

Overview

This homework extends the lecture samples (Sample 3 MCP / Sample 4 MCP Architecture) with a FastMCP HTTP server and a LiteLLM ReAct client. The server exposes three distinct tools. The client discovers those tools at runtime over Streamable HTTP and uses ReAct reasoning to call them. The repository is public so the submission can be inspected without a private-access request.

MCP tools

Tool	Purpose
lookup_inventory	SQLite lookup against inventory.db (products table)
compute_tiered_discount	Tiered discount formula based on order volume
append_audit_event	Append a timestamped event to logs/audit.log

The audit file is also published as the MCP resource file://audit/events.

Discount brackets

- 1-9 units: 0%
- 10-49 units: 5%
- 50-99 units: 12%
- 100-249 units: 18%
- 250+ units: 25%

Environment setup

Requires Python 3.11+, uv, a local LiteLLM proxy, and SQLite (stdlib). Copy `.env.example` to `.env` and set `LITELLM_API_BASE`, `LITELLM_KEY`, `LITELLM_DEFAULT_MODEL`, and `MCP_SERVER_URL`.

```
python -m pip install uv
python -m uv sync
copy .env.example .env
```

The first server start creates `inventory.db` if it is missing and seeds the products table (id INTEGER PRIMARY KEY, name TEXT NOT NULL, quantity INTEGER NOT NULL).

Run

Terminal 1 – MCP server:

```
python -m uv run python server.py
```

Terminal 2 – ReAct client (LiteLLM):

```
python -m uv run python client.py
```

The demo prompt looks up Hydraulic Pump, computes an 80-unit discount at \$120, and writes an audit event. Captured output is in `logs/execution.log`.

Proof of execution

Screenshots in required order: MCP server, ReAct client, then the local audit log.

```
(base) C:\Users\marku\Documents\DiBSE\ISE_IndustrialComputing\Hausübungen\Homework - Part 1>python -m uv run python server.py

FastMCP
FastMCP 4.0.3
https://gofastmcp.com

Server: Math_Database_Tools, 4.0.3
Deploy free: https://horizon.prefect.io

[09/05/26 19:08:26] INFO: Starting MCP server 'Math_Database_Tools' with transport 'http' on http://0.0.0.0:8080/mcp transport.py:363
INFO: Started server process [15656]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
INFO: 127.0.0.1:60150 - "POST /mcp HTTP/1.1" 200 OK
INFO: 127.0.0.1:60154 - "GET /mcp HTTP/1.1" 200 OK
INFO: 127.0.0.1:60153 - "POST /mcp HTTP/1.1" 202 Accepted
INFO: 127.0.0.1:60156 - "POST /mcp HTTP/1.1" 200 OK
INFO: 127.0.0.1:60107 - "POST /mcp HTTP/1.1" 200 OK
INFO: 127.0.0.1:53814 - "POST /mcp HTTP/1.1" 200 OK
INFO: 127.0.0.1:53817 - "POST /mcp HTTP/1.1" 200 OK
INFO: 127.0.0.1:53819 - "DELETE /mcp HTTP/1.1" 200 OK
```

Figure 1. MCP server (server.py) – FastMCP Math_Database_Tools listening on http://0.0.0.0:8000/mcp and accepting client requests.

```
(base) C:\Users\marku\Documents\DiBSE\ISE_IndustrialComputing\Hausübungen\Homework - Part 1>python -m uv run python client.py
Connecting to MCP Server at: http://localhost:8080/mcp...
Connected!
Discovered Tools: ['lookup_inventory', 'compute_tiered_discount', 'append_audit_event']

[Reasoning] Querying LLM via LiteLLM...
[Action] Executing MCP Tool 'lookup_inventory' with arguments: {'identifier': 'Hydraulic Pump'}
[Observation] Result from MCP Server: {'found': true, 'identifier': 'Hydraulic Pump', 'products': [{'id': 3, 'name': 'Hydraulic Pump', 'quantity': 35}]}

[Reasoning] Querying LLM via LiteLLM...
[Action] Executing MCP Tool 'compute_tiered_discount' with arguments: {'order_volume': 80, 'unit_price': 120}
[Observation] Result from MCP Server: {'order_volume': 80, 'unit_price': 120.0, 'bracket': '50-99', 'discount_rate': 0.12, 'subtotal': 9600.0, 'discount_amount': 1152.0, 'total': 8448.0}

[Reasoning] Querying LLM via LiteLLM...
[Action] Executing MCP Tool 'append_audit_event' with arguments: {'event': 'Quote prepared for 80 Hydraulic Pumps at $120 each. Stock quantity: 35 available. Total after 12% discount: $8448.'}
[Observation] Result from MCP Server: {'status': 'logged', 'path': 'C:\\Users\\marku\\Documents\\DiBSE\\ISE_IndustrialComputing\\Haus\\u00fcbungen\\Homework - Part 1\\logs\\audit.log', 'entry': '2026-09-05T17:09:35Z | Quote prepared for 80 Hydraulic Pumps at $120 each. Stock quantity: 35 available. Total after 12% discount: $8448.'}

[Reasoning] Querying LLM via LiteLLM...

--- Final Synthesis Result ---
The stock quantity for Hydraulic Pumps is 35, which is insufficient to fulfill the order of 80 units.

If we compute the tiered volume discount for an order of 80 units at $120 each:
- **Subtotal**: $9,600.00
- **Discount Rate**: 12%
- **Discount Amount**: $1,152.00
- **Total after Discount**: $8,448.00

Additionally, the audit log event has been successfully logged with the following entry:
"Quote prepared for 80 Hydraulic Pumps at $120 each. Stock quantity: 35 available. Total after 12% discount: $8,448."

If you need further assistance or want to adjust the quantities, please let me know!

(base) C:\Users\marku\Documents\DiBSE\ISE_IndustrialComputing\Hausübungen\Homework - Part 1>
```

Figure 2. ReAct client (client.py) – discovers lookup_inventory, compute_tiered_discount and append_audit_event, then calls all three tools.

```
(base) C:\Users\marku\Documents\DiBSE\ISE_IndustrialComputing\Hausübungen\Homework - Part 1>logs>type audit.log
2026-09-05T16:57:55Z | Prepared a quote for 80 Hydraulic Pumps at $120 each. Stock available: 35 units. Discount applied: 12% (total discount $1152). Final amount: $8448.
2026-09-05T17:09:35Z | Quote prepared for 80 Hydraulic Pumps at $120 each. Stock quantity: 35 available. Total after 12% discount: $8448.
```

Figure 3. Audit log (logs/audit.log) – timestamped events written by append_audit_event, including stock (35), 12% discount and \$8448 total.